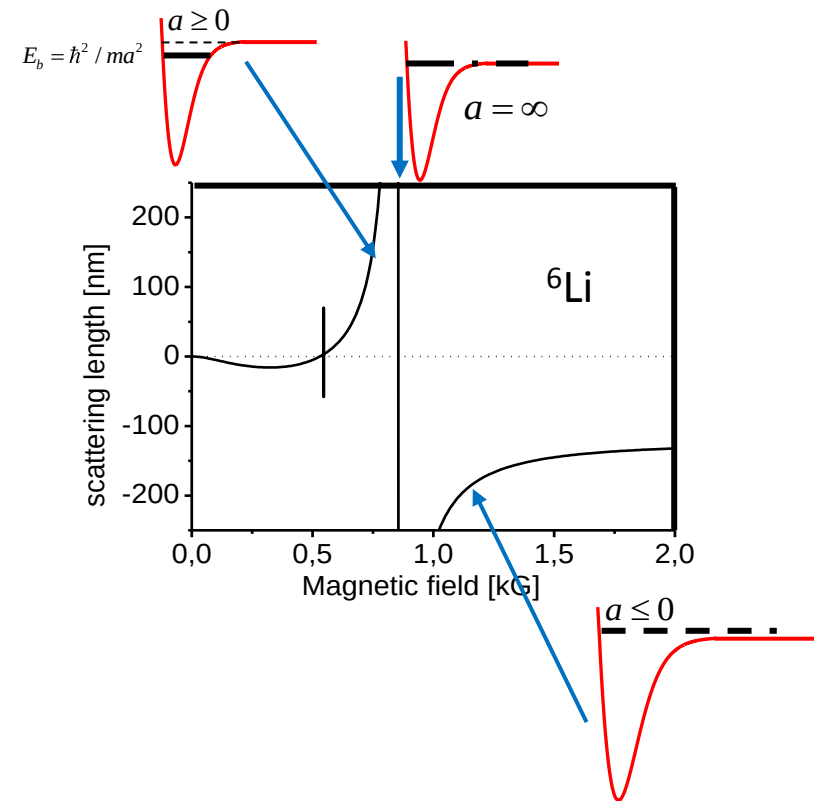


Summary of the previous lecture

- In ultra-cold atoms, interactions can be described by a contact pseudo-potential characterized by a **scattering length** a that can be controlled by an external magnetic field using **Feshbach resonances**.
- The divergence of a corresponds to the appearance of a shallow bound state on the $a > 0$ side of the resonance.
- Spin polarized fermions do not interact at low temperature.
- The density profile of the cloud in the trap can be described using the **local density approximation (LDA)**



Mean field approximation for bosons

$$\hat{H} = \int d^3\mathbf{r} \hat{\Psi}(\mathbf{r})^\dagger h_0 \hat{\Psi}(\mathbf{r}) + \frac{1}{2} \int d^3\mathbf{r} d^3\mathbf{r}' V(\mathbf{r} - \mathbf{r}') \hat{\Psi}(\mathbf{r})^\dagger \hat{\Psi}(\mathbf{r}')^\dagger \hat{\Psi}(\mathbf{r}') \hat{\Psi}(\mathbf{r})$$

For a homogeneous system, in the grand canonical ensemble :

$$h_0 = -\frac{\hbar^2 \nabla^2}{2m} - \mu$$

Mean-field approximation : the matter field is in a coherent state.

$$\hat{\Psi}(\mathbf{r}) \simeq \alpha \in \mathbb{C} \quad |\alpha|^2 = \text{density}$$

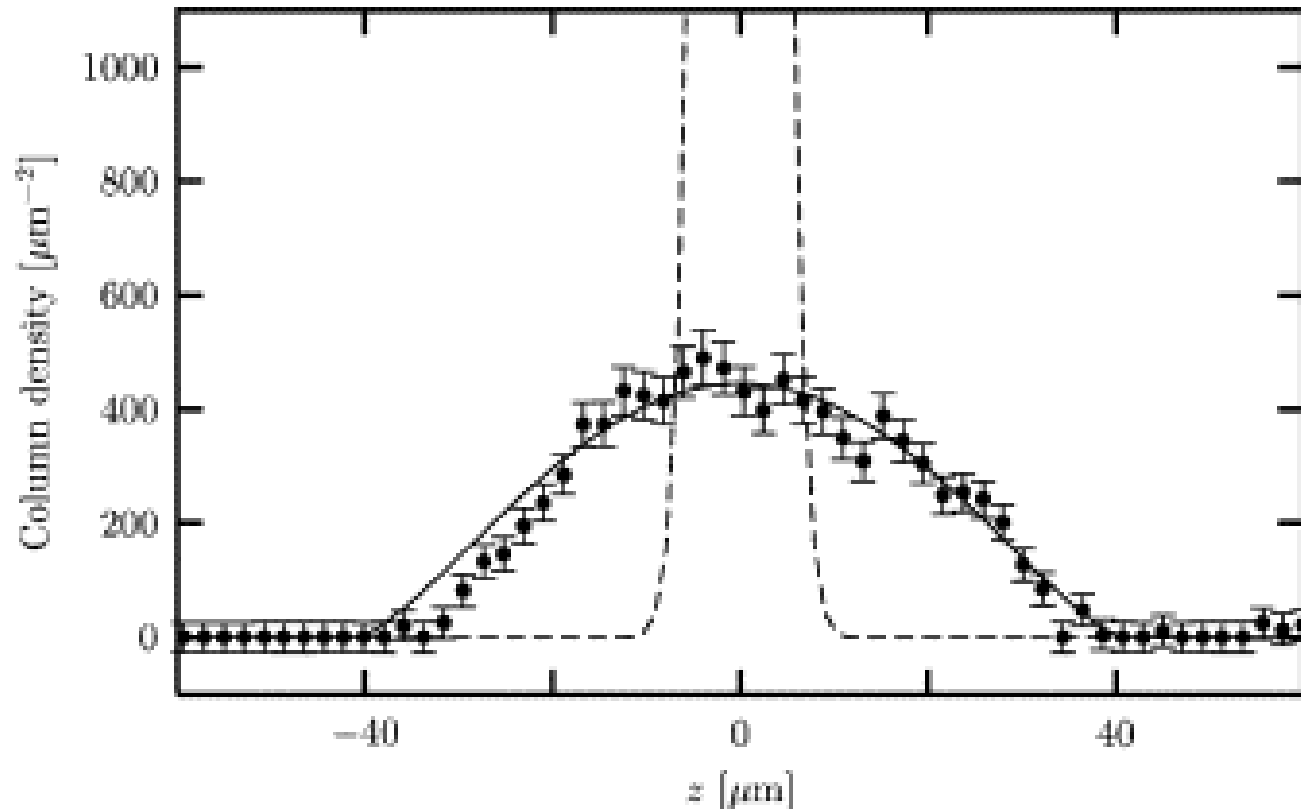
$$\text{Pseudo-potential: } H[\alpha] \simeq N \left[-\mu |\alpha|^2 + \frac{g |\alpha|^4}{2} \right]$$

$$\text{Ground state: } \frac{\partial H}{\partial \alpha^*} = \frac{\partial H}{\partial \alpha} = 0$$

Mean-field chemical potential of the uniform Bose gas : $\mu = gn$

Local density Approximation and comparison with experiments

Local Density approximation : $n(\mathbf{r}) = \frac{\mu_0 - V(\mathbf{r})}{g}$ (Thomas-Fermi profile)



Lee-Huang Yang corrections

Energy of the ground state (Lee-Huang-Yang formula, 1957)

$$E = E_{\text{Mean Field}} \left(1 + \frac{128}{15\sqrt{\pi}} \sqrt{na^3} + 8 \frac{4\pi - 3\sqrt{3}}{3} na^3 \ln(Cna^3) + \dots \right)$$

Up to this order the equation of state of the dilute Bose gas is universal (it depends only on the scattering length and not on the details of the potential).

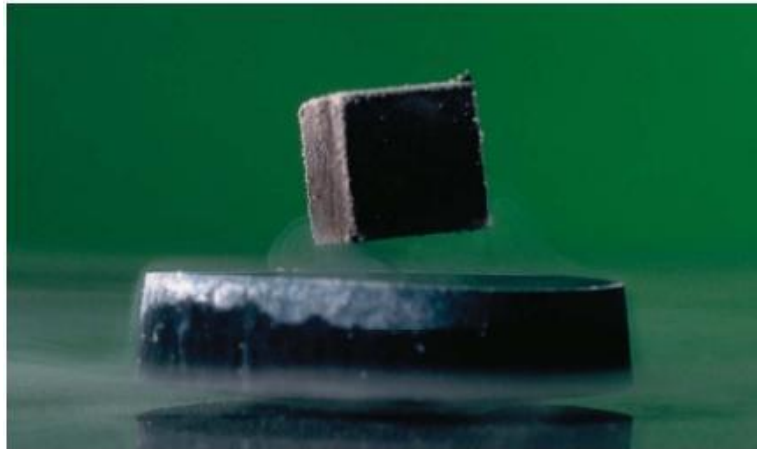
But its not true anymore beyon Lee-Huang-Yang (C depends on the details of the potentials – effective range and three-body interactions)

Fermionic superfluids

BEYOND THE BCS MODEL

BCS IN A NUTSHELL (1957):

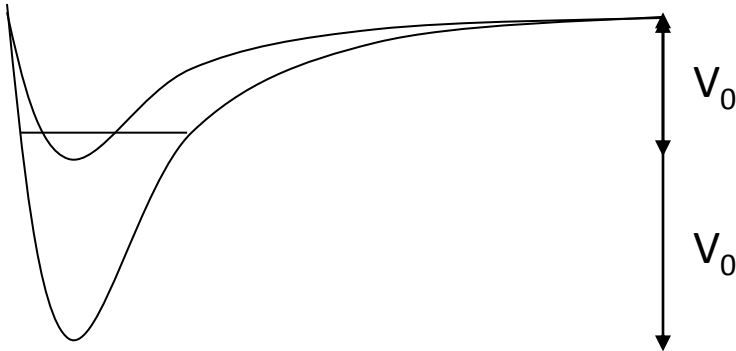
Superconductivity arises from pairing of *weakly* interacting (identical) *electrons* with *opposite spins*



WHAT IF?

- Interactions are strongly attractive (BEC/BCS crossover – Nozière, Schmitt-Rink, Leggett)?
- Spin populations are imbalanced (Clogston-Chandrasekhar limit)?
- Partners of the Cooper pair are different (exciton Bose-Einstein condensates)?
- Reduced dimensionality (1D/2D, high T_c superconductors)?

FERMIONIC SUPERFLUIDITY



DEEP POTENTIAL : stable 2-body bound state with binding energy $E_b = -\hbar^2/ma^2$. Many-body ground state: BEC of molecules ($a > 0$)

SHALLOW POTENTIAL ($V_0 < V_0^*$, $a < 0$):
No 2-body bound state, but Cooper pairing stabilized by Pauli blocking.

BOSE ENSTEIN CONDENSATION AND BCS THEORY APPEAR AS TWO LIMITING CASES OF A UNIFYING THEORY: THE BEC-BCS CROSSOVER (Eagles, Leggett, Nozière, Schmitt-Rink)

THE BCS MEAN-FIELD THEORY

Many-body hamiltonian

$$\begin{aligned}\hat{H} &= \int d^3\mathbf{r} \sum_{\sigma} \left(\hat{\Psi}_{\sigma}^{\dagger}(\mathbf{r}) h_0 \hat{\Psi}_{\sigma}(\mathbf{r}) \right) + \int d^3\mathbf{r} d^3\mathbf{r}' V(\mathbf{r} - \mathbf{r}') \hat{\Psi}_{\uparrow}^{\dagger}(\mathbf{r}) \hat{\Psi}_{\downarrow}^{\dagger}(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \hat{\Psi}_{\downarrow}(\mathbf{r}') \hat{\Psi}_{\uparrow}(\mathbf{r}) \\ &= \int d^3\mathbf{r} \sum_{\sigma} \left(\hat{\Psi}_{\sigma}^{\dagger}(\mathbf{r}) h_0 \hat{\Psi}_{\sigma}(\mathbf{r}) \right) + g_b \int d^3\mathbf{r} \hat{\Psi}_{\uparrow}^{\dagger}(\mathbf{r}) \hat{\Psi}_{\downarrow}^{\dagger}(\mathbf{r}) \hat{\Psi}_{\downarrow}(\mathbf{r}) \hat{\Psi}_{\uparrow}(\mathbf{r})\end{aligned}$$

$$h_0 = -\frac{\hbar^2}{2m} \nabla^2 - \mu \quad V(\mathbf{r}) = g_b \delta(\mathbf{r})$$

In momentum space: $\hat{\Psi}_{\sigma}(\mathbf{r}) = \sum_{\mathbf{k}\sigma} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{L^3}} \hat{a}_{\mathbf{k},\sigma}$

$$\hat{H} = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}} \hat{a}_{\mathbf{k}\sigma}^{\dagger} \hat{a}_{\mathbf{k}\sigma} + \frac{g_b}{L^3} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} \hat{a}_{\mathbf{k}+\mathbf{q}\uparrow}^{\dagger} \hat{a}_{\mathbf{k}'-\mathbf{q}\downarrow}^{\dagger} \hat{a}_{\mathbf{k}'\downarrow} \hat{a}_{\mathbf{k}\uparrow} \quad \xi_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} - \mu$$

THE BCS ORDER PARAMETER

BCS order parameter

$\chi_{\mathbf{k}} = \langle \hat{a}_{-\mathbf{k}\downarrow} \hat{a}_{\mathbf{k}\uparrow} \rangle$ ~ **order parameter** for a gas of bosonic molecules with zero center of mass momentum.

BCS hamiltonian: expand H around the coherent wave-function

$$\hat{H} = \text{cte} + \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} \hat{a}_{\mathbf{k}\sigma}^+ \hat{a}_{\mathbf{k}\sigma} + \Delta^* \sum_{\mathbf{k}} \hat{a}_{-\mathbf{k}\downarrow} \hat{a}_{\mathbf{k}\uparrow} + \Delta \sum_{\mathbf{k}} \hat{a}_{\mathbf{k}\uparrow}^+ \hat{a}_{-\mathbf{k}\downarrow}^+ + \dots$$

$$\Delta = \frac{g_b}{L^3} \sum_{\mathbf{k}} \chi_{\mathbf{k}} = \frac{g_b}{L^3} \sum_{\mathbf{k}} \langle \hat{a}_{-\mathbf{k}\downarrow} \hat{a}_{\mathbf{k}\uparrow} \rangle$$

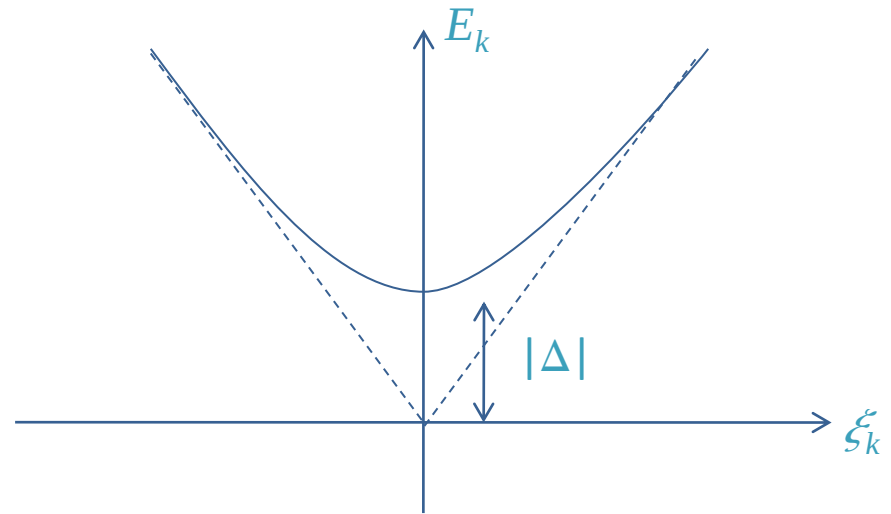
$$\hat{H} = \text{cte} + \sum_{\mathbf{k}} \begin{pmatrix} \hat{a}_{\mathbf{k}\uparrow} \\ \hat{a}_{-\mathbf{k}\downarrow}^+ \end{pmatrix}^+ \cdot \begin{pmatrix} \xi_{\mathbf{k}} & \Delta \\ \Delta^* & -\xi_{\mathbf{k}} \end{pmatrix} \cdot \begin{pmatrix} \hat{a}_{\mathbf{k}\uparrow} \\ \hat{a}_{-\mathbf{k}\downarrow}^+ \end{pmatrix} + \dots$$

BOGOLIUBOV TRANSFORMATION FOR FERMIONS

$$\begin{pmatrix} \xi_k & \Delta \\ \Delta^* & -\xi_k \end{pmatrix} \begin{pmatrix} u_k \\ v_k \end{pmatrix} = E_k \begin{pmatrix} u_k \\ v_k \end{pmatrix}$$

$$|v_k|^2 = 1 - |u_k|^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right)$$

$$E_k = \sqrt{\xi_k^2 + |\Delta|^2}$$



Define the **fermionic** annihilation operators α_k and β_k

$$\begin{cases} \hat{\alpha}_k = u_k \hat{a}_{k\uparrow} + v_k \hat{a}_{-k\downarrow}^+ \\ \hat{\beta}_{-k}^+ = -v_k \hat{a}_{k\uparrow} + u_k \hat{a}_{-k\downarrow}^+ \end{cases} \Rightarrow \hat{H} = \text{cte} + \sum_k E_k \left(\hat{\alpha}_k^+ \hat{\alpha}_k + \hat{\beta}_k^+ \hat{\beta}_k \right) + \dots$$

The BCS ground state is the vacuum of the Bogoliubov excitations
 $|\Delta|$ is the excitation gap

GAP EQUATION

Self-consistency gap equations

$$\Delta = -\frac{g_b}{L^3} \sum_{\mathbf{k}} u_{\mathbf{k}} v_{\mathbf{k}} = -g_b \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{\Delta}{2\sqrt{|\Delta|^2 + \xi_{\mathbf{k}}^2}}$$

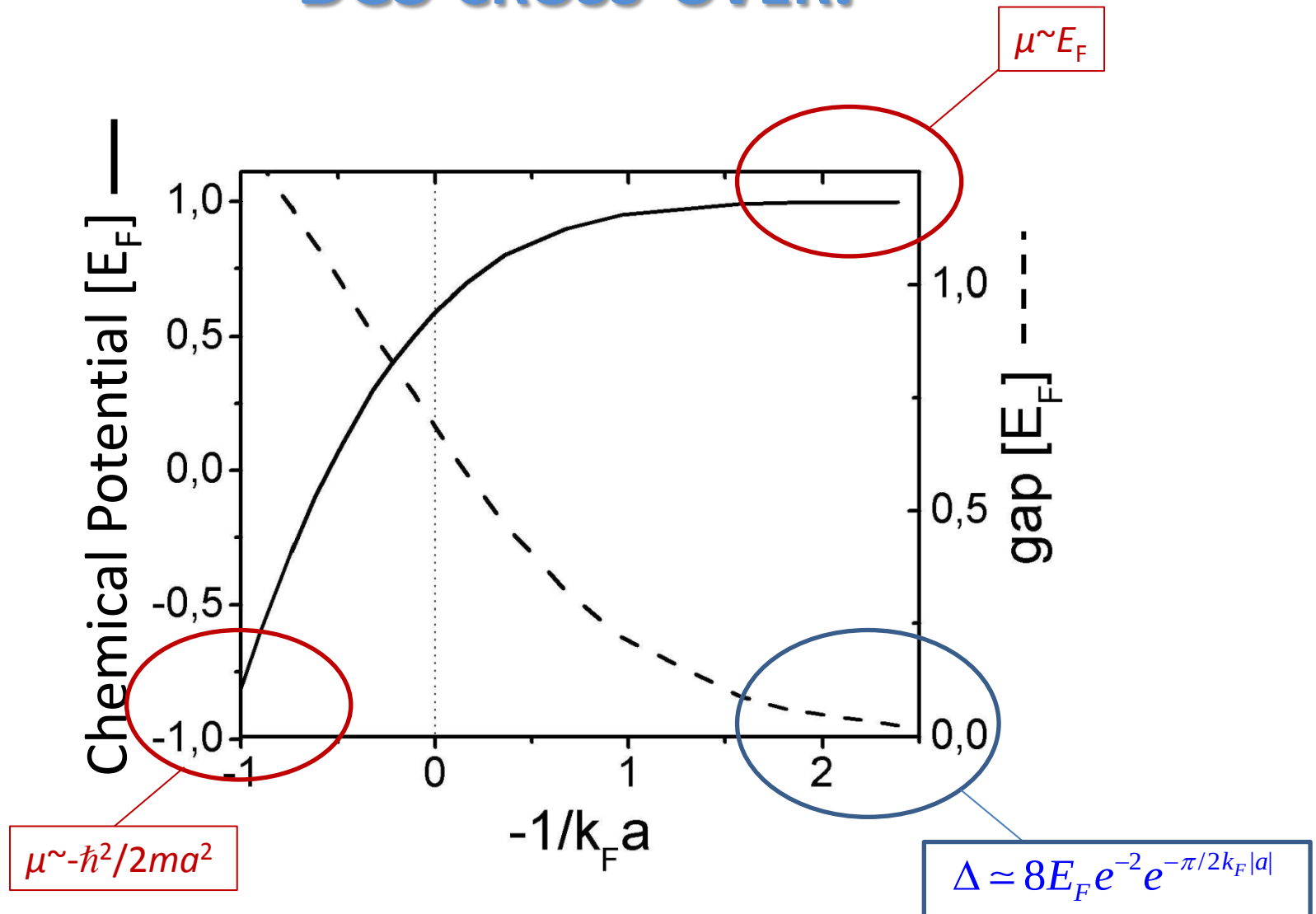
The sum diverges! **Use a regularisation of the coupling constant**

$$\frac{1}{g_b} = \frac{1}{g} - \frac{1}{L^3} \sum_{\mathbf{k}} \frac{m}{\hbar^2 k^2} \Rightarrow \frac{1}{g} = - \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \left[\frac{1}{2\sqrt{|\Delta|^2 + \xi_{\mathbf{k}}^2}} - \frac{m}{\hbar^2 k^2} \right] \quad \text{Gap Equation}$$

Compute the density to get the chemical potential:

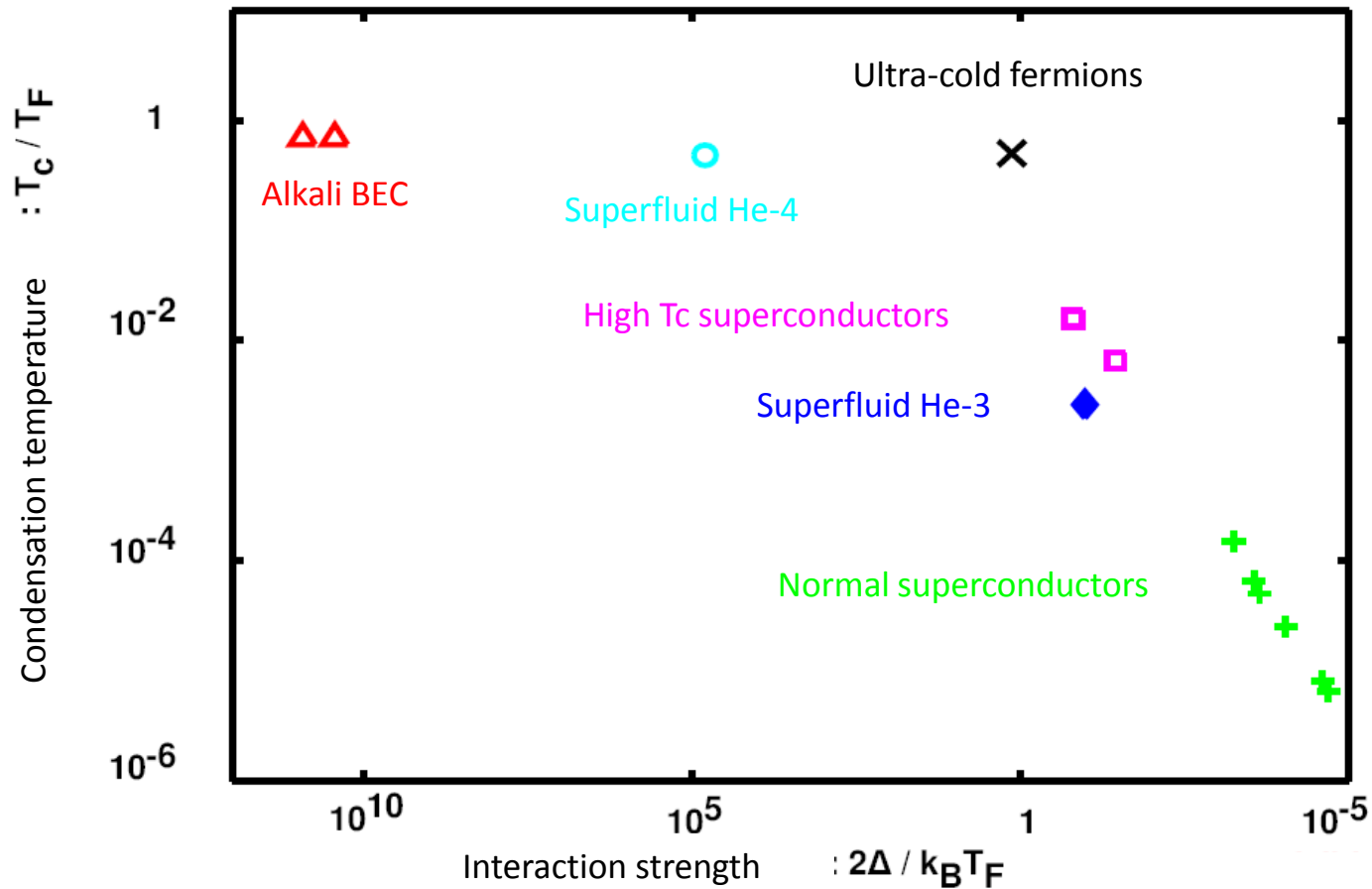
$$n_{\uparrow} = \frac{1}{L^3} \sum_{\mathbf{k}} \langle \hat{a}_{\mathbf{k}\uparrow}^{\dagger} \hat{a}_{\mathbf{k},\downarrow} \rangle = \frac{1}{L^3} \sum_{\mathbf{k}} v_{\mathbf{k}}^2 = \frac{1}{2} \int \frac{d^3 k}{(2\pi^3)} \left(1 - \frac{\xi_k}{E_k} \right)$$

THE SUPERCONDUCTING GAP IN THE BEC-BCS CROSS-OVER.



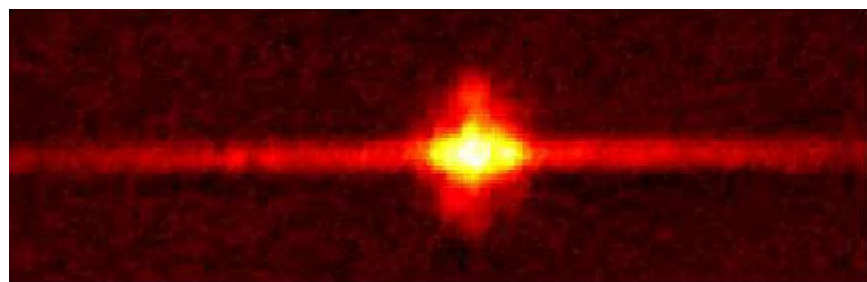
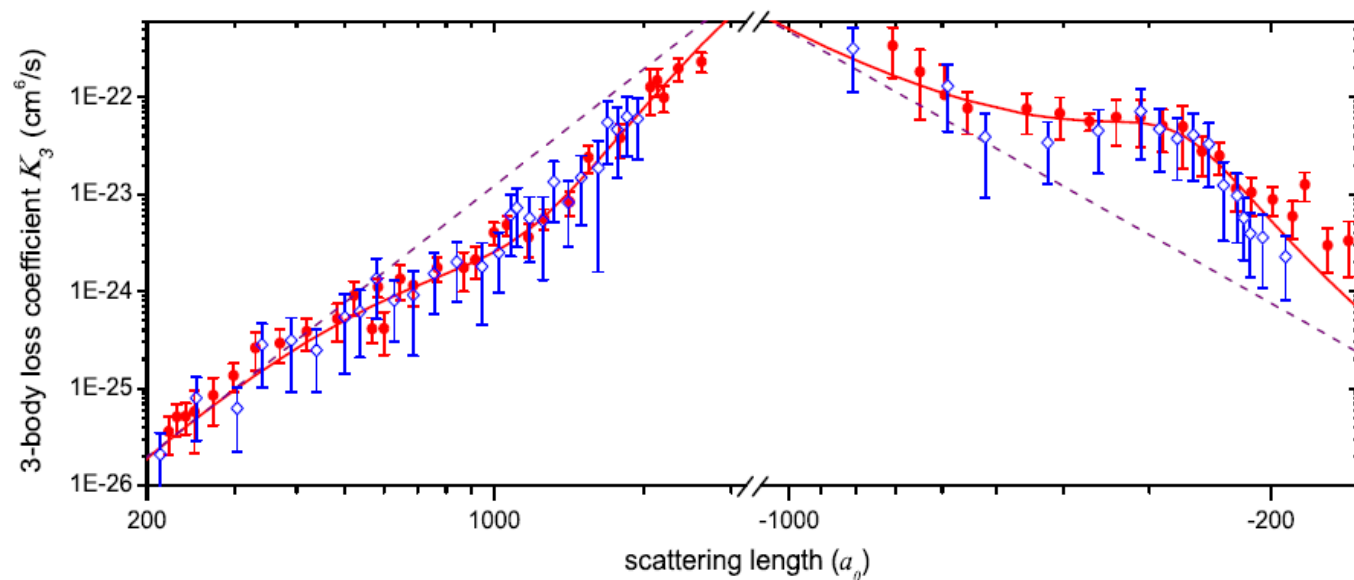
THE EXPERIMENTAL EXPLORATION OF THE BEC-BCS CROSSOVER

THE BEC-BCS PATCHWORK



FESHBACH RESONANCES WITH BOSONS

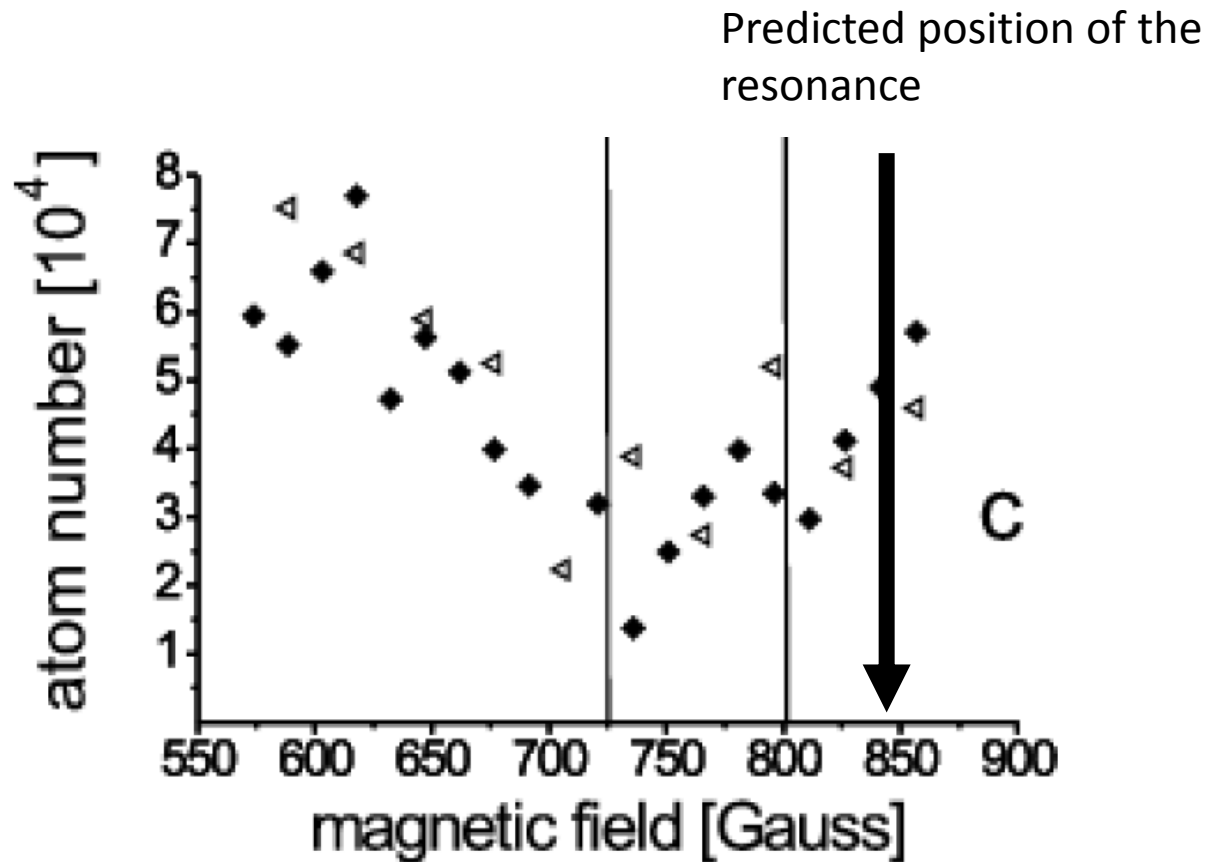
$$L_3 \propto a^4$$



Bose-Nova

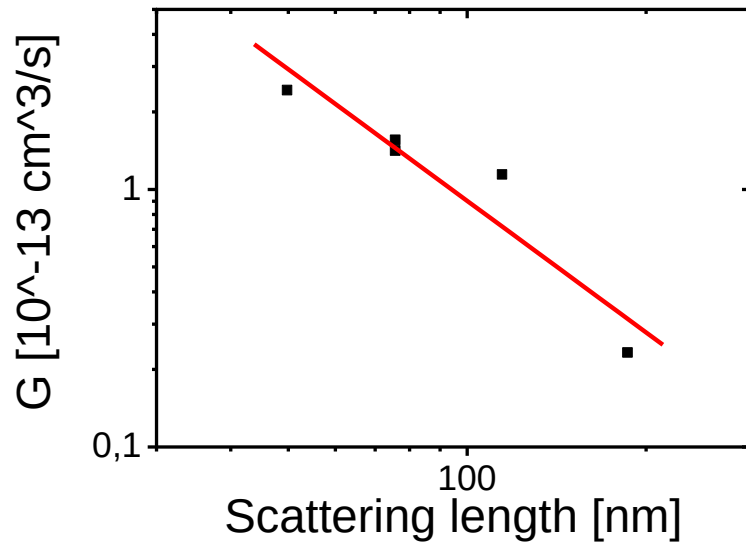


LOOKING FOR THE FESHBACH RESONANCE OF LITHIUM 6



Losses in ${}^6\text{Li}$ (fermions)

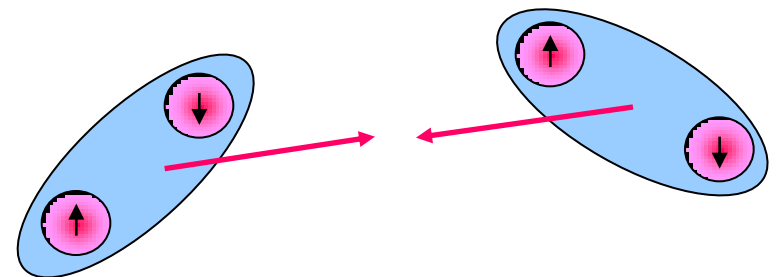
INHIBITION OF INELASTIC LOSSES



Scaling law

$$G \sim a^{-2.0 \pm 0.8} \quad (\text{theory Petrov } et al. G \sim a^{-2.55})$$

3/4 atom loss requires 2 atoms of same spin close to each other.

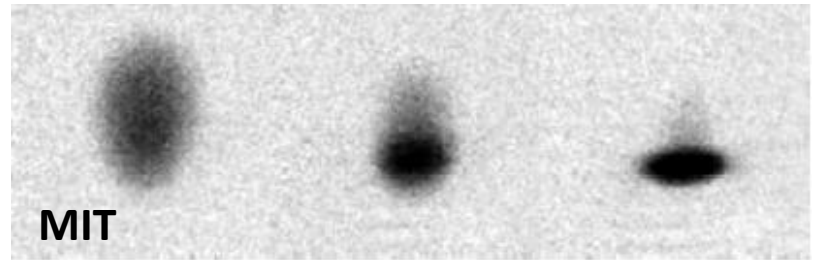
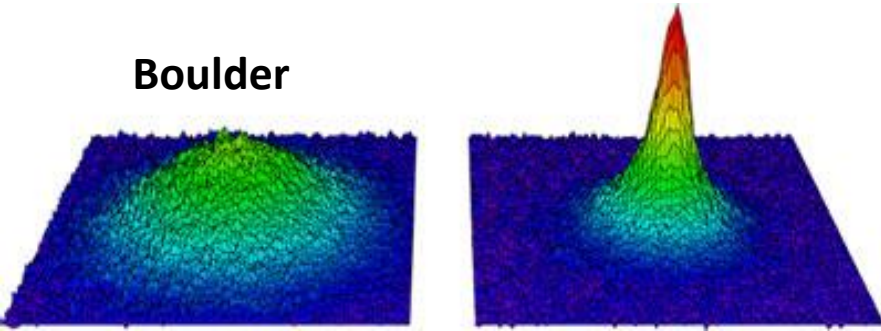


2 body (dimers) losses mainly : decay towards deeply bound states)

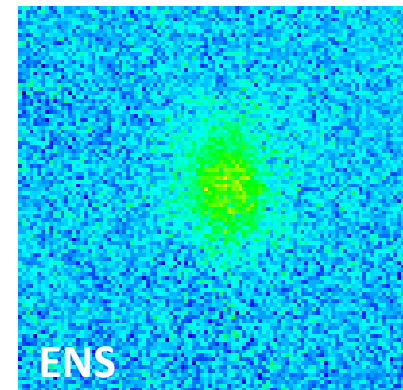
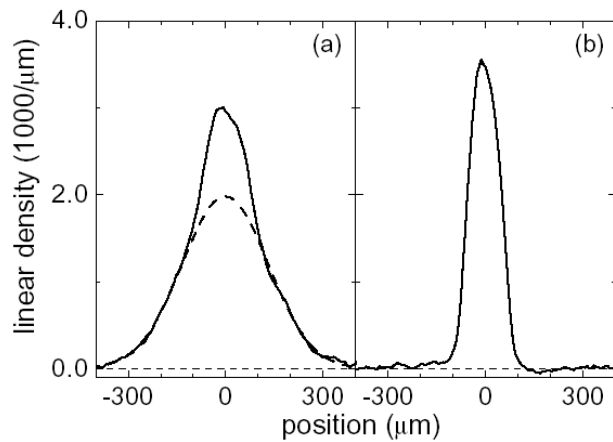
$$\dot{N} = -G \langle n \rangle N$$

2003-2013: TEN YEARS OF STRONGLY CORRELATED FERMION GASES

Boulder



Innsbruck



Also at Rice, Duke, Tokyo, Melbourne...

LOCAL THERMODYNAMICS OF TRAPPED GASES

$$\bar{n}(z) = \int dx dy n(x, y, z)$$

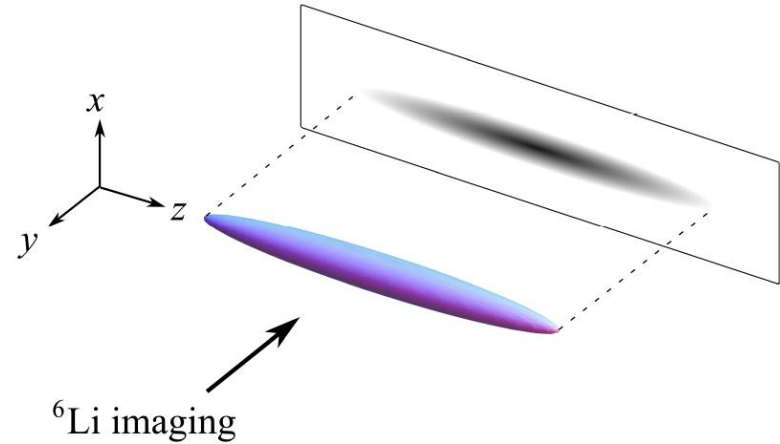
•Cylindrical trap and harmonic trapping:
 $dx dy = 2\pi r dr = -2\pi d\mu / m\omega^2$

•Gibbs-Duhem $n = dP/d\mu$

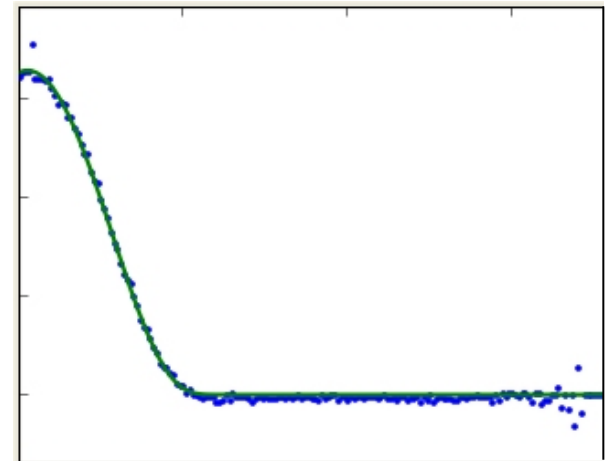
$$\bar{n}(z) = - \int \frac{2\pi d\mu}{m\omega^2} \frac{\partial P}{\partial \mu} = \frac{2\pi P(\mu(z), T)}{m\omega^2}$$

High accuracy (few percent):

- Double integration increases S/N ratio
- One shot yields a whole piece of the Equation of state



« P » ~~$\bar{n}$~~



~~z~~ « μ »

THE GROUND STATE OF A HOMOGENEOUS FERMİ GAS

Our goal : measure the Equation of State of the homogeneous Fermi gas

$$\begin{aligned}\Omega(T, V, \mu) &= E - TS - \mu N \\ &= -P(\mu, T)V\end{aligned}$$

Pressure contains all the thermodynamic information

In this talk : Low temperature limit : $T = 0$

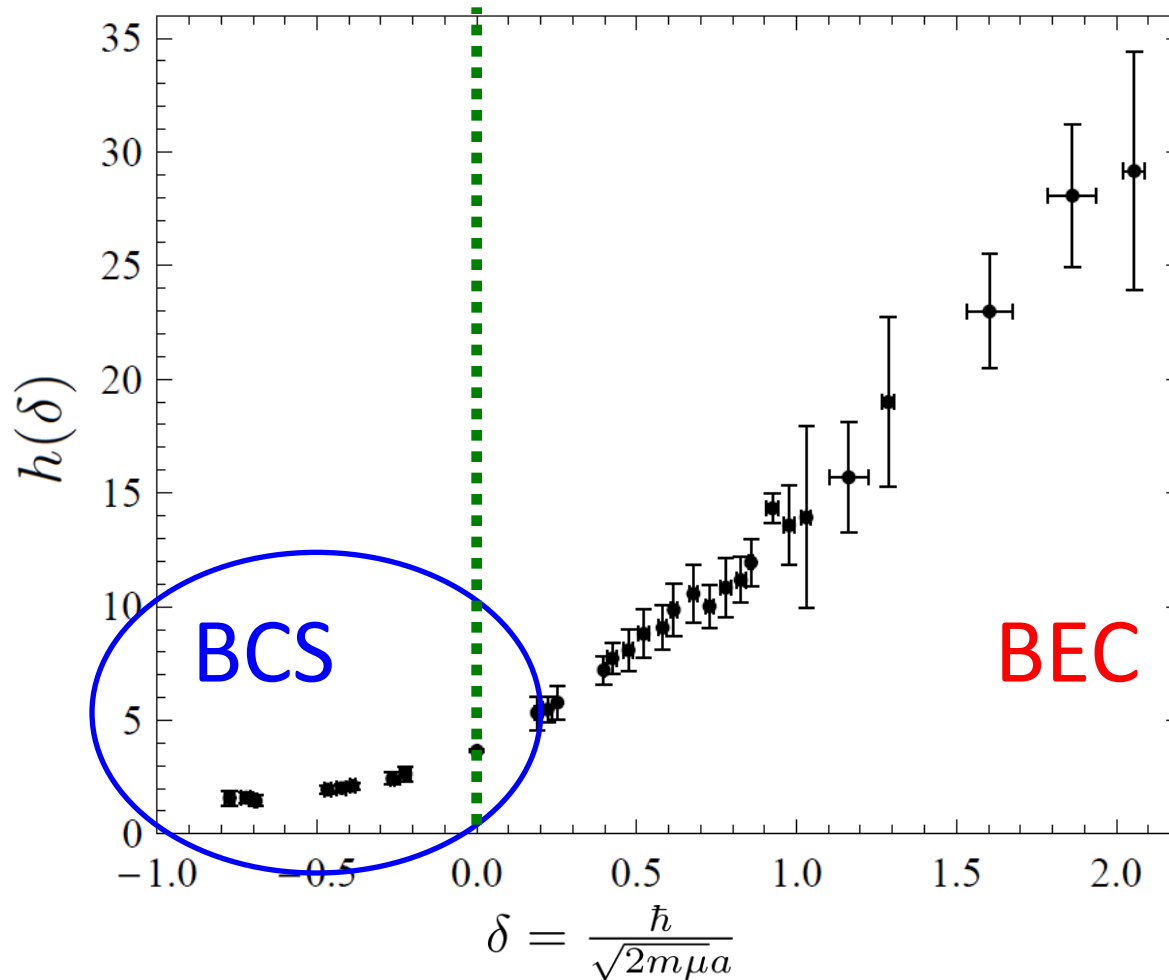
Universal function

Normalized pressure : $P(\mu) = 2P_0(\mu)h(\delta)$

Fermi pressure of an ideal gas $P_0(\mu, T = 0) = \frac{1}{15\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \mu^{5/2}$

Interaction parameter $\delta = \frac{\hbar}{\sqrt{2m\tilde{\mu}a}}$ with $\tilde{\mu} = \mu - \Theta(a)\frac{E_b}{2}$
grand-canonical analog to $\frac{1}{k_F a}$

THE EOS OF A FERMIONIC SUPERFLUID WITH TUNABLE INTERACTIONS AT (ALMOST) ZERO TEMPERATURE

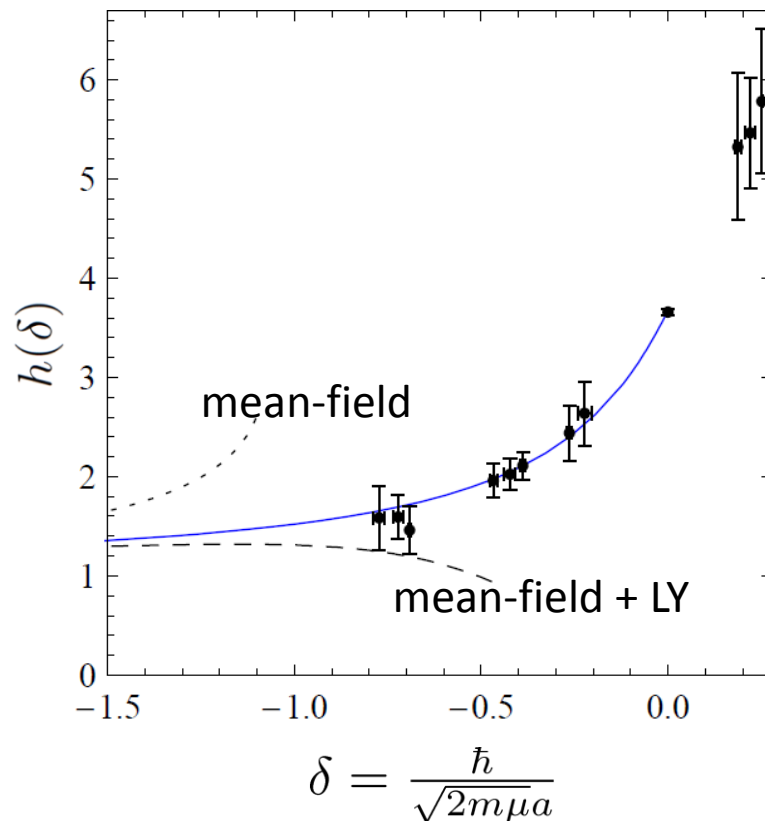


ASYMPTOTICS I: BCS REGIME

$$E = \frac{3}{5} N E_F \left(1 + \frac{10}{9\pi} k_F a + \frac{4(11 - 2 \log 2)}{21\pi^2} (k_F a)^2 \dots \right)$$

MF

Lee-Yang



Using Padé interpolation,
we find : 0.18(2)

LY calculation : $\frac{4(11 - 2 \log 2)}{21\pi^2} \simeq 0.186$

T.D. Lee, C.N. Yang, Phys. Rev. 105, 1119 (1957)

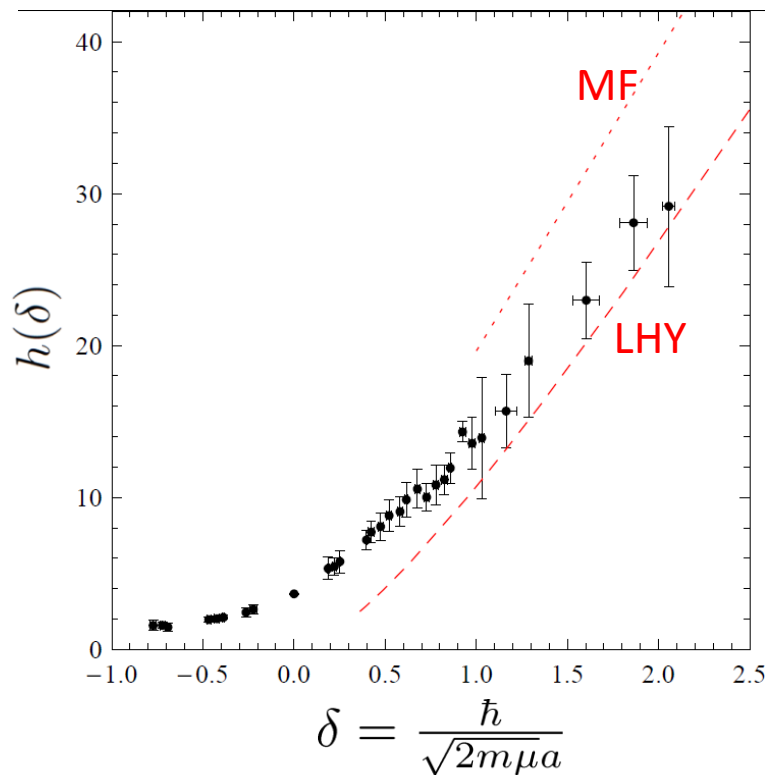
ASYMPTOTICS II: BEC REGIME

$$E = \frac{N}{2}E_b + N \frac{\pi \hbar^2 a_{dd}}{2m} n \left(1 + \frac{128}{15\sqrt{\pi}} \sqrt{na_{dd}^3} + \dots \right)$$

Binding Energy

MF

Lee-Huang-Yang



We find : 4.4(5)

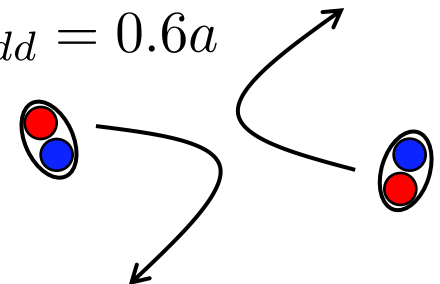
LHY calculation : $\frac{128}{15\sqrt{\pi}} \simeq 4.81$

T.D. Lee, K. Huang, C.N. Yang, Phys. Rev. 106, 1135 (1957)

No effect of composite nature of dimers
(except a_{dd})

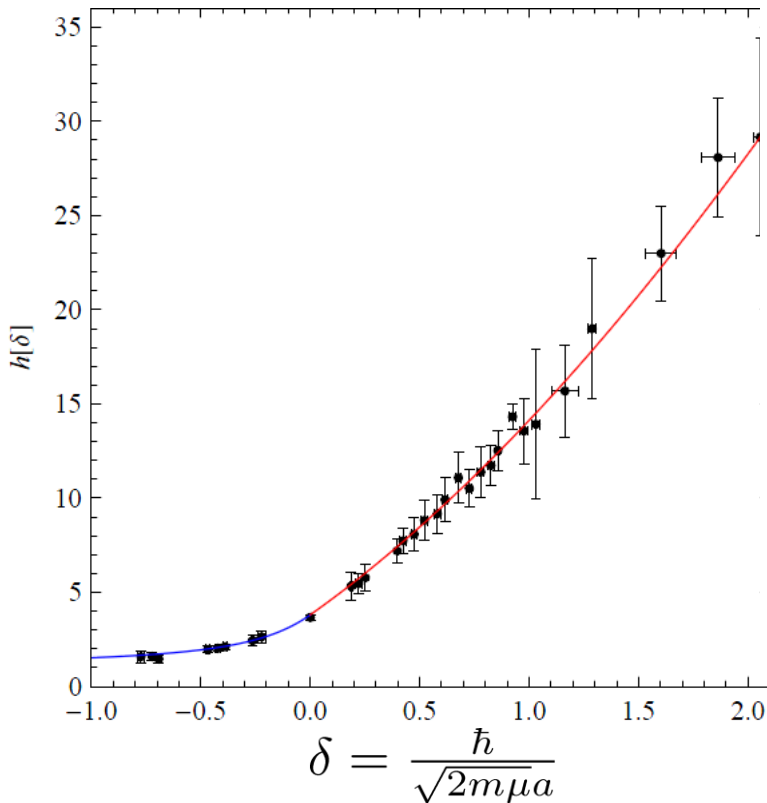
D. Petrov et al. PRL (2004), X. Leyronas *et al*, PRL (2007)

$$a_{dd} = 0.6a$$



ASYMPTOTICS III: UNITARITY

$$E = \frac{3}{5} N E_F \left(\xi_s - \zeta \frac{1}{k_F a} + \dots \right)$$



$\mu = \xi_s E_F$ Superfluid T=0 Bertsch's parameter

$$\xi_s = 0.39(1)$$

In agreement with MIT's value

S. Tan's contact constant

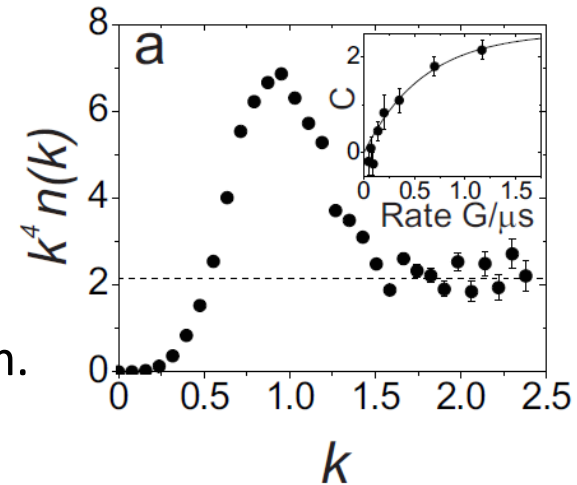
$$\zeta = 0.93(5)$$

TAN'S CONTACT PARAMETER

At large momenta, the momentum distribution of the gas scales like

$$n_k \underset{k \rightarrow \infty}{\sim} \frac{C}{k^4}$$

C characterizes **the short range correlations** of the system.



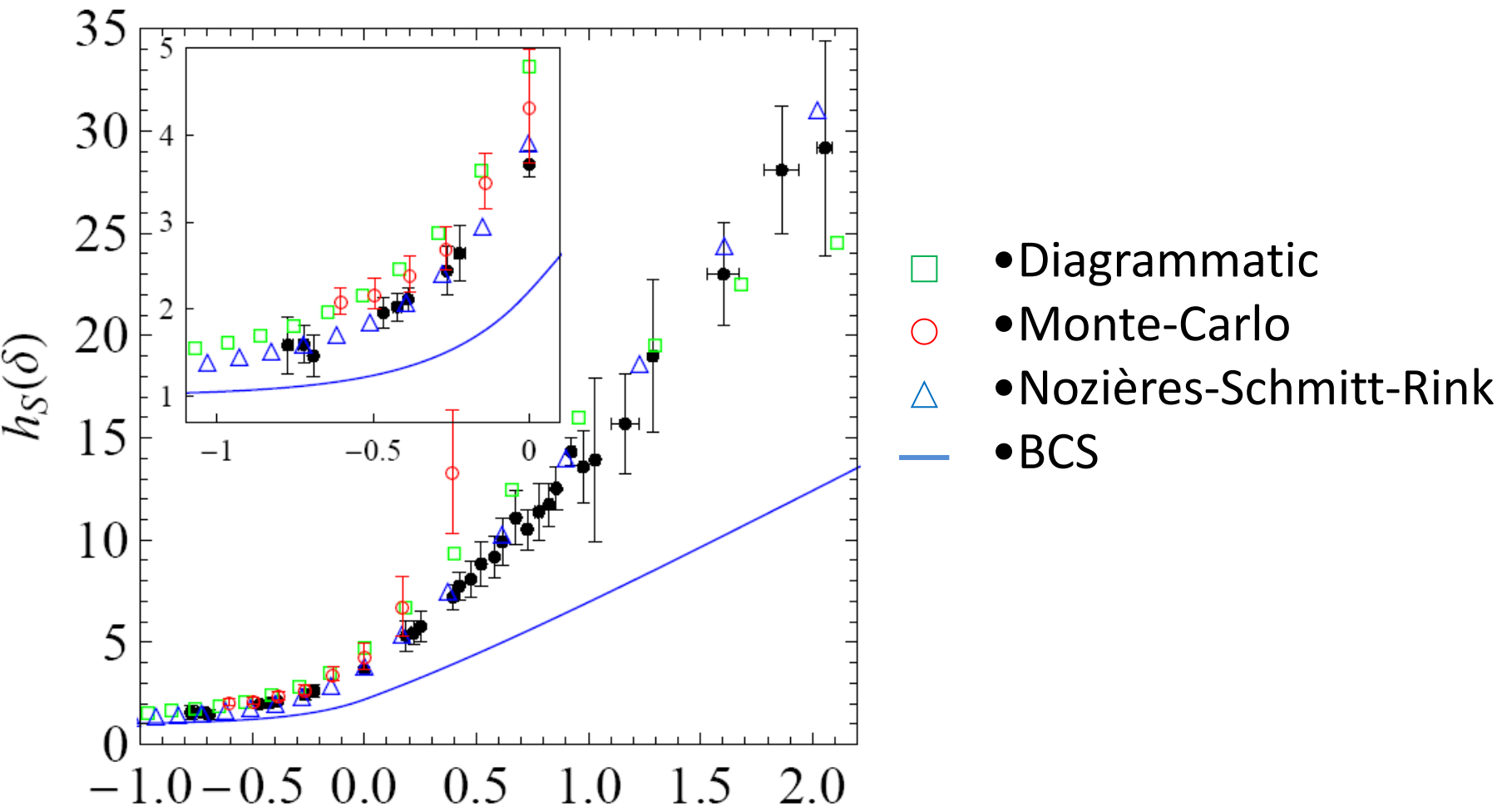
JILA : PRL **104**, 235301 (2010)

Tan's sdiabatic theorem: C is related to the equation of state of the gas.

$$n_k \underset{k \rightarrow \infty}{\sim} - \frac{1}{2V \varepsilon_k^2} \frac{dE}{d(1/g)}$$

NB : the contact parameter can be measured by rf spectroscopy (JILA) or Bragg spectroscopy (Swinburne, Kunhle *et al.*, PRL **105**, 070402 (2010))

COMPARISON WITH ADVANCED SIMULATION METHODS



FINITE TEMPERATURE

FINITE TEMPERATURE BCS THEORY

Bogoliubov modes are populated according to Fermi-Dirac distribution

$$f(E) = \frac{1}{1 + e^{E/k_B T}}$$

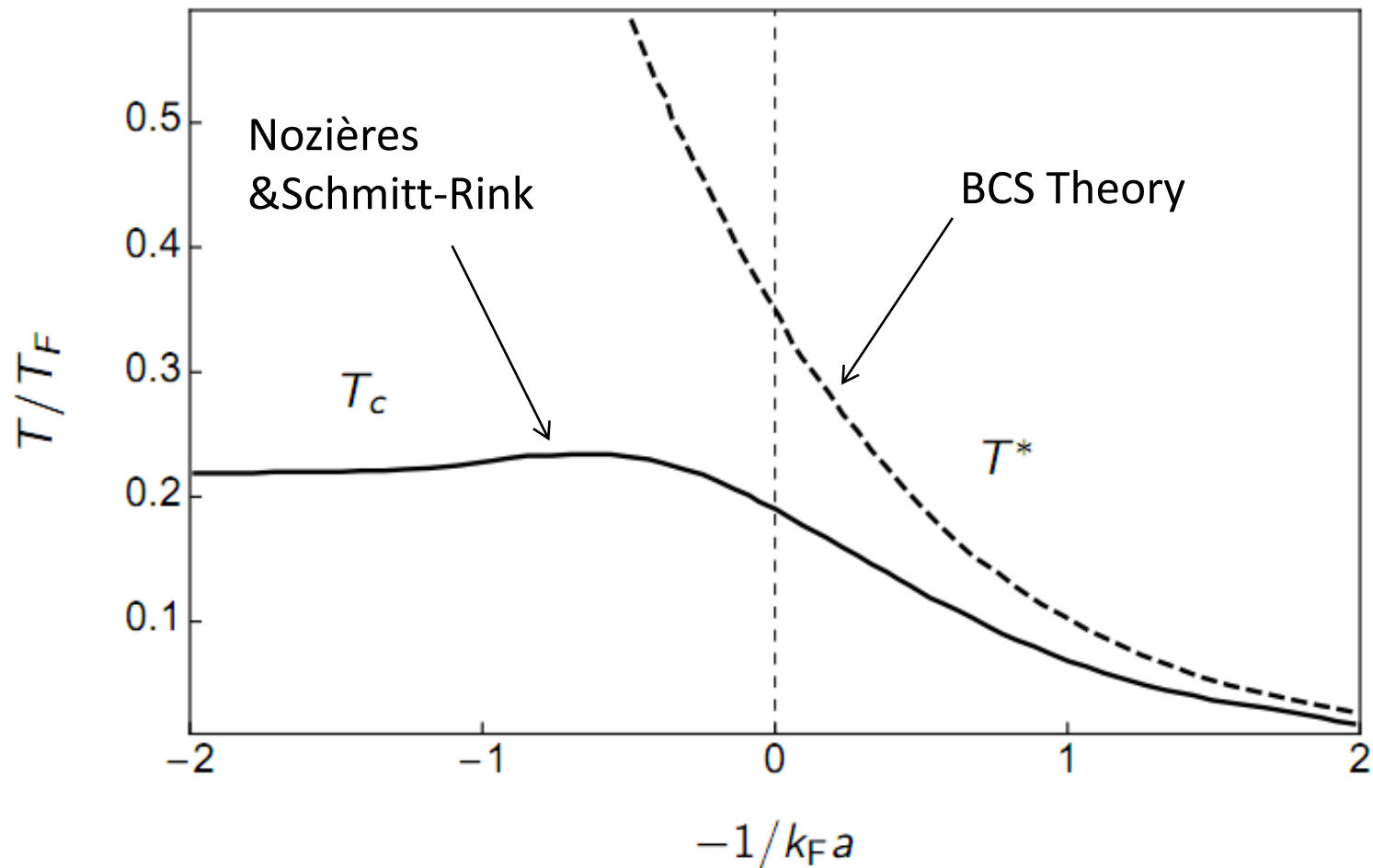
$$\langle \hat{a}_{-\mathbf{k}\downarrow} \hat{a}_{\mathbf{k}\uparrow} \rangle = -u_{\mathbf{k}} v_{\mathbf{k}} \left(1 - \langle \hat{\alpha}_{\mathbf{k}}^+ \hat{\alpha}_{\mathbf{k}} \rangle - \langle \hat{\beta}_{-\mathbf{k}}^+ \hat{\beta}_{-\mathbf{k}} \rangle \right)$$

Gap equation

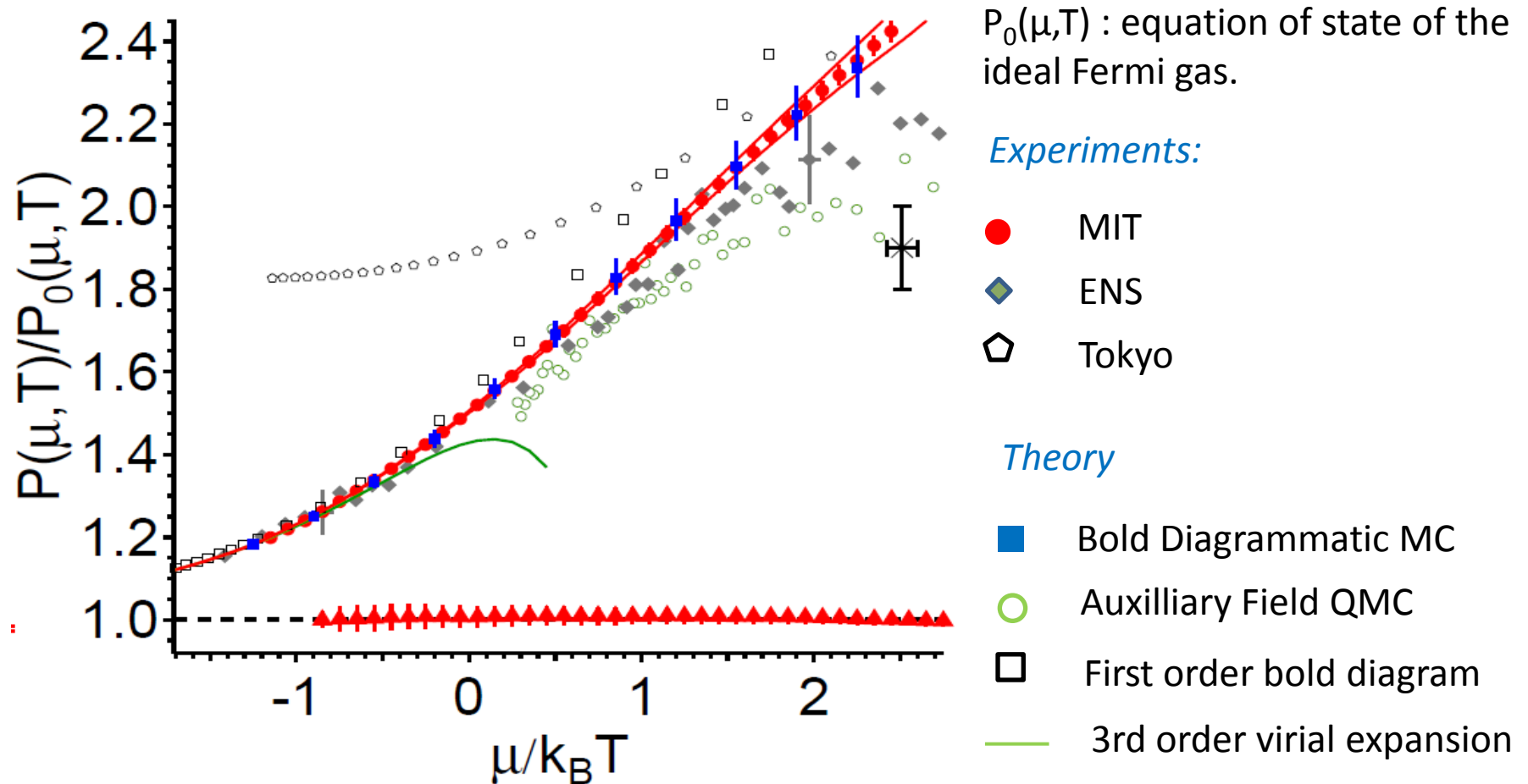
$$\Delta = \frac{g_b}{L^3} \sum_{\mathbf{k}} \langle \hat{a}_{-\mathbf{k}\downarrow} \hat{a}_{\mathbf{k}\uparrow} \rangle$$

$$\frac{1}{g} = - \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \left[\frac{\tan(E_{\mathbf{k}} / 2k_B T)}{2E_{\mathbf{k}}} - \frac{m}{\hbar^2 k^2} \right] \quad \Delta(T_c) = 0$$

THE CRITICAL TEMPERATURE OF A FERMIONIC SUPERFLUID



THE EQUATION OF STATE OF THE FINITE TEMPERATURE FERMI GAS



HIGH TEMPERATURE EXPANSION

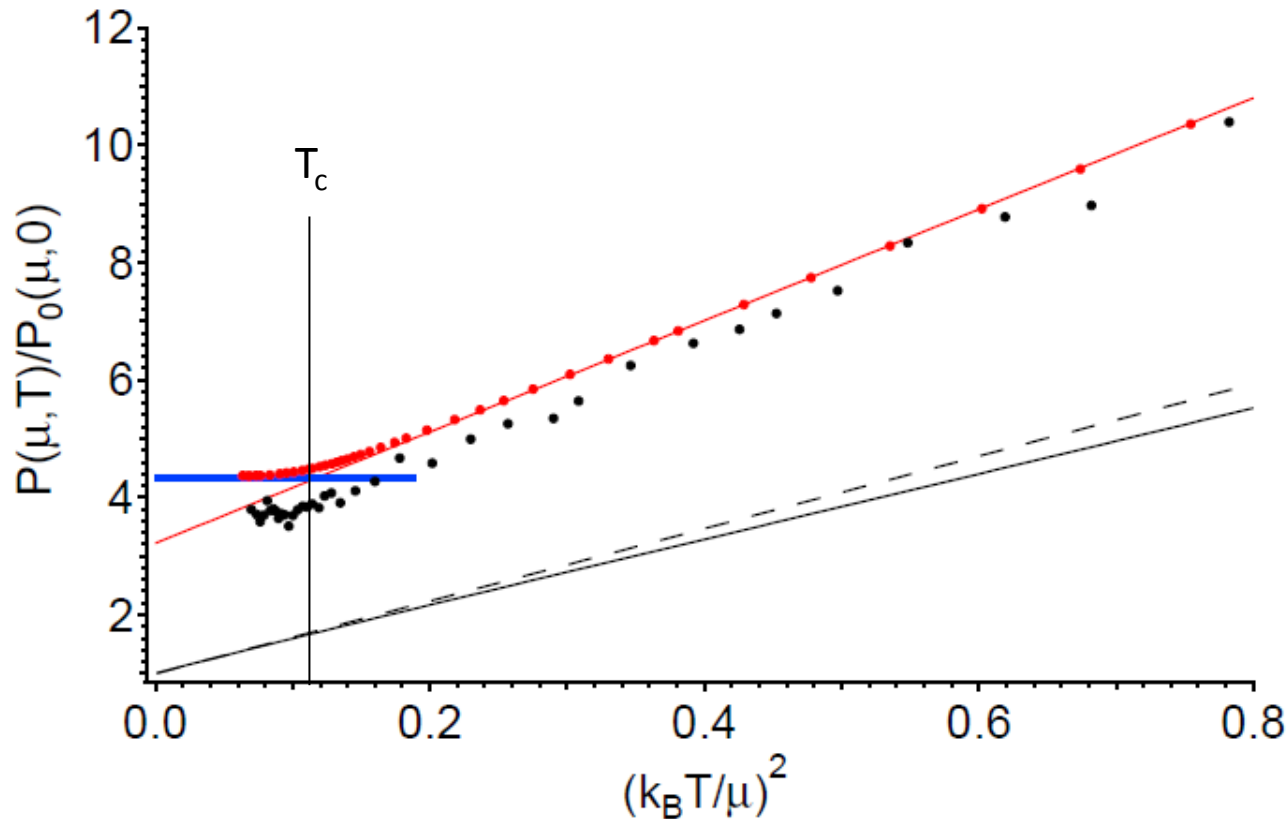
High temperature limit: Virial expansion

$$P = 2 \frac{k_B T}{\lambda_{\text{th}}^3} \sum_n b_n e^{n\beta\mu}$$

Calculating b_n requires solving the n-body problem

	Theory	Exp.
b_2	$3 \times 2^{1/2} / 8 \sim 0.53$	0.53
b_3	-0.29 (Drummond)	-0.29
	1.11 (Rupak)	
b_4	-0.016 (Blume)	0.065

LOW TEMPERATURE BEHAVIOR



$$T_c/T_F \sim 0.15$$

$$T < T_c$$

$$P(\mu, 0) = P_0(\mu, 0) \xi_s^{-3/2}$$

$$\begin{aligned} \xi_s &= 0.376(5) \text{ (MIT)} \\ &= 0.39(1) \text{ (ENS)} \end{aligned}$$

$T > T_c$: Fermi Liquid behavior (specific heat $\sim T$)

$$P(\mu, T) = P_0(\mu, 0) \left(\xi_n^{-3/2} + \frac{5\pi^2}{8} \xi_n^{-1/2} \frac{m^*}{m} \left(\frac{k_B T}{\mu} \right)^2 \right)$$

CURRENT OPEN PROBLEMS

- **Nature of the normal phase at unitarity.** Is it really a Fermi Liquid(evidences for the contrary at JILA)? Is its structure related to the exotic normal phases of the high-Tc superconductors.
- **Dynamics of strongly correlated systems :**
 - ☐ Observation of second sound and measurement of the superfluid fraction (Innsbruck)
 - ☐ Observation of solitons (MIT), but debated : the effective mass is too large and some studies suggest it might be a vortex ring.
 - ☐ Observation of the fermionic Anderson-Higgs mode : oscillations of the order parameter after an abrupt change of the scattering length (to be done).
- **Fermions in optical lattices**
 - ☐ Mott transition (Zurich, Munich)
 - ☐ Observation of antiferromagnetic correlations (Zurich, Rice ?)
 - ☐ Observation of a superfluid phase in the repulsive Hubbard model.

...